

A Fejér Four-Sign Gate for H3

Sliding-Window TT^* , Diagonal Feasibility, and a Non-Circular Sufficient Reduction on the Complete Actual Block

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Abstract

TPC-108 gives the exact full-range TT^* identity for one literal generic affine Möbius block, but its off-diagonal contains four Möbius values. We turn that observation into a quantitative, truncated, all-prefix certificate. For a block

$$S = \sum_{z=1}^N A_z e^{-2\pi i \alpha z}, \quad C(h) = \sum_z A_{z+h} \overline{A_z},$$

sliding-window Cauchy–Schwarz gives, for $1 \leq H \leq N$,

$$|S|^2 \leq \frac{N+H-1}{H} \sum_{|h|<H} \left(1 - \frac{|h|}{H}\right) e^{-2\pi i \alpha h} C(h).$$

The Fejér sum is real and nonnegative. Writing $V = \sum |A_z|$, $E = \sum |A_z|^2$, and $\mathfrak{p} = V^2/(NE)$, we obtain

$$\frac{|S|^2}{V^2} \leq 2 \left\{ \frac{1}{H\mathfrak{p}} + \frac{N}{HV^2} (\mathcal{O}_H)_+ \right\},$$

where $\mathcal{O}_H$ is the signed Fejér off-diagonal. In the TPC-127 shift-two frame its arithmetic factor is exactly

$$\lambda(n+qh-2)\lambda(n+qh)\lambda(n-2)\lambda(n),$$

with every shifted squarefree, coprimality, periodic, physical, and prefix factor retained. This yields a precise sufficient four-sign gate and an exponent ledger. It also proves a sharp diagonal stop: sparse support with $H\mathfrak{p} = O(1)$ can saturate the original sum even when every short off-diagonal vanishes. The finite inequality is L0, its complete literal crosswalk is L1, and the required growing four-sign estimate remains unproved L2 input.

1 The finite sliding-window inequality

Let $A_1, \dots, A_N \in \mathbb{C}$, extend $A_z = 0$ outside $[1, N]$, and define

$$S(\alpha) = \sum_{z=1}^N A_z e^{-2\pi i \alpha z}, \tag{1}$$

$$C(h) = \sum_{z \in \mathbb{Z}} A_{z+h} \overline{A_z}. \tag{2}$$

Thus $C(-h) = \overline{C(h)}$.

Theorem 1.1 (Sliding-window Fejér TT^*). *For every integer $H \geq 1$,*

$$\boxed{|S(\alpha)|^2 \leq \frac{N+H-1}{H} \mathcal{K}_H(\alpha)}, \tag{3}$$

where

$$\mathcal{K}_H(\alpha) = \sum_{|h| < H} \left(1 - \frac{|h|}{H}\right) e^{-2\pi i \alpha h} C(h) \geq 0. \quad (4)$$

The identity behind the inequality is

$$H\mathcal{K}_H(\alpha) = \sum_{t=2-H}^N \left| \sum_{j=0}^{H-1} A_{t+j} e^{-2\pi i \alpha(t+j)} \right|^2. \quad (5)$$

Proof. Put $b_z = A_z e^{-2\pi i \alpha z}$, extended by zero, and

$$B_t = \sum_{j=0}^{H-1} b_{t+j}.$$

Every b_z occurs in exactly H of the $N + H - 1$ windows, so

$$\sum_{t=2-H}^N B_t = HS(\alpha).$$

Cauchy–Schwarz gives

$$H^2 |S(\alpha)|^2 \leq (N + H - 1) \sum_t |B_t|^2.$$

Expanding the last square and grouping by $h = j - k$ yields

$$\sum_t |B_t|^2 = \sum_{|h| < H} (H - |h|) e^{-2\pi i \alpha h} C(h),$$

which is (5) and proves the theorem. $\square$

Remark 1.2 (Relation to TPC-108). Taking all differences gives the exact full TT^* identity of TPC-108 [2]. [Theorem 1.1](#) instead inserts a predeclared short shift scale H , positive Fejér weights, and the precise Cauchy cost needed for a sufficient estimate. The finite Fejér/van der Corput device is classical [1, Chapter 8]; the new content here is its literal fixed-two four-sign crosswalk and audited all-prefix stop ledger.

2 Diagonal participation and the signed off-diagonal

Put

$$V = \sum_{z=1}^N |A_z|, \quad E = \sum_{z=1}^N |A_z|^2. \quad (6)$$

If $E > 0$, define the participation ratio

$$\mathfrak{p} = \frac{V^2}{NE} \in (0, 1]. \quad (7)$$

The upper bound is Cauchy–Schwarz. If $V = 0$, then every $A_z = 0$, so $E = S = 0$; such a prefix is discharged trivially and omitted from normalized maxima. On every remaining prefix $V > 0$, hence $E > 0$ and (7) is defined.

Write

$$\mathcal{K}_H(\alpha) = E + \mathcal{O}_H(\alpha), \quad (8)$$

where

$$\mathcal{O}_H(\alpha) = 2 \operatorname{Re} \sum_{h=1}^{H-1} \left(1 - \frac{h}{H}\right) e^{-2\pi i \alpha h} C(h). \quad (9)$$

Corollary 2.1 (Normalized Fejér certificate). *If $1 \leq H \leq N$ and $V > 0$, then*

$$\boxed{\frac{|S(\alpha)|^2}{V^2} \leq 2 \left\{ \frac{1}{H\mathfrak{p}} + \frac{N}{HV^2} (\mathcal{O}_H(\alpha))_+ \right\}}. \quad (10)$$

Proof. Since $N + H - 1 \leq 2N$, [theorem 1.1](#) gives

$$\frac{|S|^2}{V^2} \leq \frac{2N}{HV^2} (E + \mathcal{O}_H).$$

Use $NE/V^2 = 1/\mathfrak{p}$ and replace the off-diagonal by its positive part. $\square$

Definition 2.2 (Typed four-sign certificate). For exponents $\eta_{\text{diag}}, \eta_4 \geq 0$, a block-prefix and a shift scale $H \leq N$ pass the Fejér certificate if

$$\frac{1}{H\mathfrak{p}} \leq X^{-2\eta_{\text{diag}}+o(1)}, \quad (11)$$

$$\frac{N}{HV^2} (\mathcal{O}_H(\alpha))_+ \leq X^{-2\eta_4+o(1)}. \quad (12)$$

The same $o(1)$ must be uniform over the predeclared actual block-prefix family.

Theorem 2.3 (Exact exponent transfer). *If (11)–(12) hold uniformly, then*

$$|S(\alpha)| \leq V X^{-\eta_{\text{block}}+o(1)}, \quad \eta_{\text{block}} = \min\{\eta_{\text{diag}}, \eta_4\}. \quad (13)$$

Proof. Insert the two inputs in (10); fixed constants are absorbed by the uniform $X^{o(1)}$, and take square roots. $\square$

3 The complete fixed-two arithmetic crosswalk

We now identify $\mathcal{O}_H$ without suppressing its coefficient. For one actual TPC-108 block, TPC-127 gives

$$n(z) = n_0 + qz, \quad q = as,$$

and

$$A_z = \lambda(q)\epsilon_z \lambda(n(z) - 2)\lambda(n(z))\Psi_z. \quad (14)$$

Here

$$\begin{aligned} \epsilon_z &= \mu^2(D(z))\mu^2(V(z)), \\ \Psi_z &= \mathbf{1}_{(B, V(z))=1} \chi(\tau + Bz) W_X(\tau + Bz). \end{aligned} \quad (15)$$

Both factors are literal; in particular, the physical weight has not been replaced by 1 [\[5\]](#).

Proposition 3.1 (Actual four-sign kernel). *For $h \neq 0$,*

$$\begin{aligned} C(h) &= \sum_{\substack{z \in I \\ z+h \in I}} \epsilon_{z+h} \epsilon_z \lambda(n(z) + qh - 2)\lambda(n(z) + qh) \\ &\quad \times \lambda(n(z) - 2)\lambda(n(z))\Psi_{z+h} \overline{\Psi_z}. \end{aligned} \quad (16)$$

Consequently (12) is a signed Fejér average of the complete actual four-sign kernel, including the additive factor $e^{-2\pi i \alpha h}$.

Proof. Insert (14) in $C(h) = \sum A_{z+h} \overline{A_z}$. Since $\lambda(q)^2 = 1$ and $n(z+h) = n(z) + qh$, the displayed formula follows. $\square$

Remark 3.2 (TPC-128 components). One may first use the finite CRT expansion of TPC-128 and reparametrize a retained component. The same theorem then holds with the enlarged progression step $ask^2\ell^2$, while its tail and census remain separate inputs [\[6\]](#). The complete formula (16) does not require that optional truncation.

4 All prefixes and complete-family return

H3 requires the declared prefix family, not just one terminal interval. For every actual block ξ and required prefix T , form

$$A_{\xi,T}, \quad N_{\xi,T}, \quad V_{\xi,T}, \quad E_{\xi,T}, \quad \mathfrak{p}_{\xi,T}, \quad C_{\xi,T}(h)$$

by restricting the literal coefficient before applying [theorem 1.1](#). Choose $H_{\xi,T}$ by a rule fixed before estimating the correlations. Prefixes with $V_{\xi,T} = 0$ are zero and pass trivially; the normalized certificate is imposed only on the remaining prefixes.

Corollary 4.1 (Termwise H3 ledger). *Suppose every retained actual block-prefix passes [definition 2.2](#), and suppose the verified complete-family outer mass has cost $X^{\ell_{\text{out}}+o(1)}$ in the normalization of TPC-108/TPC-126. Then the termwise return exponent is*

$$\eta_{\text{net}} = \min\{\eta_{\text{diag}}, \eta_4\} - \ell_{\text{out}}. \quad (17)$$

At the local amplitude level, the square-root counterpart of a quadratic $X^{-1/200+o(1)}$ target is

$$\min\{\eta_{\text{diag}}, \eta_4\} - \ell_{\text{out}} \geq \frac{1}{400}. \quad (18)$$

The legacy 1/200 termwise linear-window screen is a deliberately stronger sufficient condition. It is not the exact quadratic-to-amplitude conversion and is not necessary for the local Fejér certificate.

Proof. [Theorem 2.3](#) gives the uniform block-prefix saving. Sum with the separately verified outer absolute-mass bound, exactly as in the TPC-108 termwise ledger. $\square$

[Corollary 4.1](#) does not identify the local mass V with the final physical quadratic normalization, nor does it supply the prefactor and synthesis crosswalk needed at the strict endpoint. Those remain separate certificates. It also does not replace the determinant/zero-mode compatibility of TPC-121/TPC-122 [\[3, 4\]](#).

5 Sharp structural stops

Proposition 5.1 (Short-off-diagonal vanishing can still saturate). *Fix $H \geq 2$. Let $A_z = 1$ on m positions separated by at least H , and $A_z = 0$ elsewhere in an interval of length $N \asymp mH$. At $\alpha = 0$,*

$$C(h) = 0 \quad (0 < |h| < H), \quad |S| = V = m,$$

while

$$H\mathfrak{p} = H \frac{m^2}{Nm} \asymp 1.$$

Thus no saving follows from vanishing short off-diagonals when the diagonal feasibility condition fails.

Proof. The support separation proves the correlation statement. The remaining identities follow from $V = E = m$ and $N \asymp mH$. $\square$

Proposition 5.2 (Generic phase alone is insufficient). *For any α , the bounded coefficient $A_z = e^{2\pi i \alpha z}$ satisfies*

$$S(\alpha) = N.$$

Hence neither $N\|\alpha\| > 1$ nor the Fejér identity gives cancellation without information about the literal coefficient.

Proof. Every summand in [\(1\)](#) equals 1. $\square$

DIAGONAL STOP. If no predeclared $H \leq N$ makes $H\mathfrak{p}$ grow at the required power on every retained prefix, this certificate cannot give that saving. Increasing H beyond the prefix length is not permitted.

CIRCULARITY STOP. The four-sign input must be proved directly from (16). Deducing it from the desired bound for S and feeding it back into theorem 2.3 is circular. For a visibly local reduction, one should predeclare $H \leq N^{1-\varepsilon}$ (or another audited sublength range); choosing H only after observing a favorable global sum is not allowed.

ABSOLUTE-VALUE STOP. The minimal gate is the signed positive part in (12). Replacing it by $\sum_{h < H} |C(h)|$ is a valid but stronger sufficient condition. Failure of that stronger condition does not close the signed Fejér route.

ARITHMETIC STOP. No theorem in this paper estimates the growing four-Liouville expression (16) with its actual progression, masks, phase, physical weights, origins, and all-prefix uniformity. That is the remaining L2 gate.

6 Verdict

Statement	Level and status
Sliding-window identity and Fejér inequality	Exact finite theorem (L0).
Participation normalization and exponent transfer	Exact deterministic certificate (L0).
Four-sign formula for the complete fixed-2 literal block	Proven crosswalk (L1).
Uniform diagonal participation on the full actual family	Must be audited; not asserted globally.
Signed four-sign estimate (12)	Not proved; new arithmetic input (L2 if proved).
H3 threshold, physical 1/400, or twin-prime theorem	Not proved.

TPC-130 advances H3 by replacing “control the TT^* off-diagonal” with one normalized, all-prefix, non-circular four-sign inequality and a separate diagonal feasibility test. The road remains open only if both tests pass on the actual family. This is a rigorous reduction, not a parity breakthrough, prime-pair lower bound, Hardy–Littlewood asymptotic, or proof of the twin-prime conjecture.

References

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